CSE7302- University Project
Review-1

Development of a Smart Blind Stick for Enhanced Mobility and Safety of Visually Impaired Persons.

Batch Number:CSE_146

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Problem Statement Number:

Organization: Presidency University

Category (Hardware / Software / Both) : Both

Problem Description:

- The proposed Smart Blind Stick will be an innovative mobility aid integrating ultrasonic sensors, infrared sensors, and GPS navigation to help visually impaired persons detect obstacles (both at ground and head level), potholes, and fast-approaching vehicles.
- It will use vibration and audio feedback to alert the user in real time. The stick will also have an emergency SOS button that can send the user's location to a registered guardian via SMS or mobile app.
- To ensure affordability and accessibility, the design will use low-cost yet durable components, rechargeable batteries, and lightweight materials. The device will be compact, easy to use, and scalable for mass production.

Objectives:

1. A low-cost Smart Blind Stick prototype with integrated sensors and feedback system
2. Mobile app connectivity for navigation and emergency support.
3. Real-time obstacle detection within a range of 2–3 meters.
4. Lightweight, foldable, and ergonomically designed for daily use.
5. Affordable solution (< ₹3000 per unit) for large-scale deployment.

Background:

According to WHO, over 39 million people worldwide are blind, and nearly 285 million have some form of visual impairment. In India, millions of visually impaired individuals face mobility challenges daily. Traditional white canes provide only limited assistance in detecting ground-level obstacles and do not alert users about overhanging objects, vehicles, or dangerous situations.

With the integration of modern technologies like sensors, GPS, and IoT, there is a need to design an affordable and smart blind stick that ensures independence, safety, and improved navigation for visually impaired individuals.

Analysis of Problem Statement

Existing System and Its Limitations

- Conventional blind sticks detect obstacles **only after physical contact**.
- They cannot identify **overhead obstacles, moving objects, water puddles, or stairs**.
- No real-time alerts or intelligent feedback mechanisms.
- Lack of emergency support and location awareness.
- Not suitable for dynamic and crowded environments.

Need for the Proposed System

- Enhances obstacle detection **before collision**.
- Provides **real-time alerts** through audio, vibration, or haptic feedback.
- Ensures user safety in both **indoor and outdoor environments**.
- Promotes **independent navigation** and confidence.

Analysis of Problem Statement (contd...)

Software and Hardware Requirements

Software:

- **Aurdino**

Hardware:

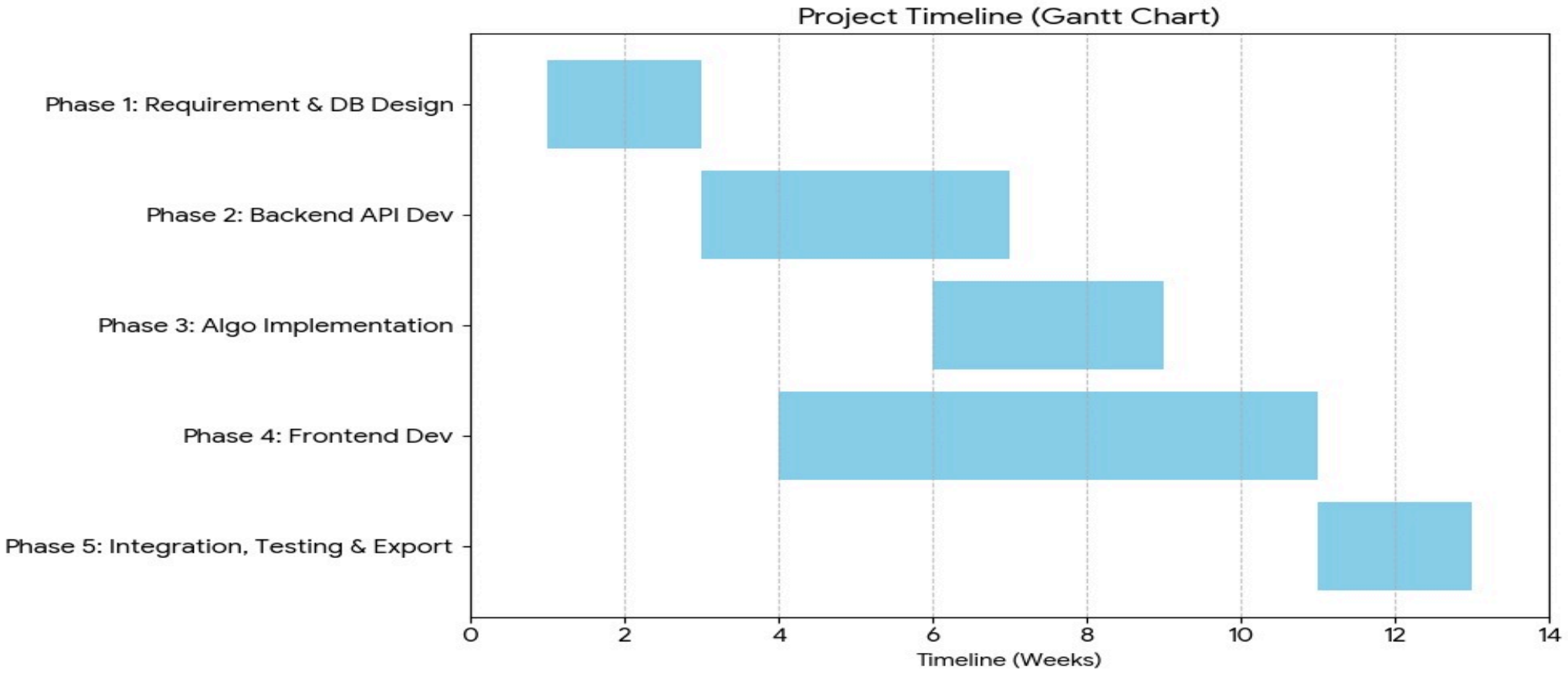
- **Batteries(CELL Rechargeable), Ultrasonic Sensor, SOS Button, A9G board, Xiao C3 Speed studio Board, Few Wires, piezoelectric diaphragm, Foam Board, Shoe, Connectors, PVC Pipe**

Github Link

Github Link

https://github.com/Ruchith04/CapstoneReview

Timeline of the Project (Gantt Chart)



References (IEEE Paper format)

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Thank
You!